Statistics/Data analysis

Title

rdhte —— RD Heterogeneous Treatment Effects Estimation and Inference.

Syntax

rdhte *depvar* *runvar* [*if*] [*in*] [**,** **covs_hte(***covars***)** **c(***#***)** **p(***#***)** **q(***#***)** **h(***#***)** **h_l(***#***)** **h_r(***#***)** **kernel(***kernelfn***)** **vce(***vcetype***)** **level(***#***)** **precision(***single*|*double***)** **covs_eff(***covars***)** **weights(***wtvar***)** **bwselect(***bwmethod***)** **bwjoint** **labels**]

Description

rdhte provides estimation and inference for heterogeneous treatment effects in RD designs using local polynomial regressions, allowing for interactions with pretreatment covariates (Calonico, Cattaneo, Farrell, Palomba and Titiunik, 2025a). Inference is implemented using robust bias-correction methods (Calonico, Cattaneo, and Titiunik, 2014)

Companion commands are: rdbwhte for data-driven bandwidth selection and rdhte_lincom for testing linear restrictions of paramaters. More general post-estimation linear hypotheses can be tested with the Stata function test.

A detailed introduction to **rdhte** in Stata is given in Calonico, Cattaneo, Farrell, Palomba and Titiunik (2025b).

Related software packages for analysis and interpretation of RD designs and related methods are available in:

https://rdpackages.github.io/

For background methodology, see Calonico, Cattaneo, Farrell, and Titiunik (2019), Calonico, Cattaneo and Farrell (2020), Cattaneo and Titiunik (2022).

Options

Estimand

c(*#***)** specifies the RD cutoff for *indepvar*. Default is **c(0)**.

covs_hte(*covars***)** specifies covariate(s) for heterogeneous treatment effects. Factor variables notation can be used to distinguish between continuous and categorical variables, select reference categories, specify interactions between variables, and include polynomials of continuous variables. If not specified, the RD Average Treatment Effect is computed.

labels displays the final RD estimates using variable labels from **covs_hte(***covars***)**.

Local Polynomial Regression

p(*#***)** specifies the order of the local polynomial used to construct the point estimator. Default is **p(1)** (local linear regression).

q(*#***)** specifies the order of the local polynomial used to construct the bias correction. Default is q(2) (local quadratic regression).

h(*#***)**, **h_l(***#***)** and **h_r(***#***)** set the bandwidths to construct the RD estimator. The same choice could be used on each side of the cutoff (via **h(***#***)**), or different to the left and right (using **h_l(***#***)** and **h_r(***#***)**). More than one bandwidth can be specified for categorical covariates. If not specified, bandwidths are computed by the companion command rdbwhte.

kernel(*kernelfn***)** specifies the kernel function used to construct the local-polynomial estimator(s). Options are: **triangular**, **epanechnikov**, and **uniform**. Default is **kernel(triangular)**.

covs_eff(*covars***)** specifies additional covariates to be used for efficiency improvements.

weights(*wtvar***)** specifies a non-negative variable used to weight the estimation procedure. The unit-specific weights multiply the kernel function.

precision(*single*|*double***)** controls the storage type used for generated working variables, including the centered running variable, polynomial terms, selected bandwidths, and kernel weights. The default is **precision(double)**. Use **precision(single)** to restore Stata's legacy float temporary-variable path for backward numerical compatibility.

Data-Driven Bandwidth Selection

bwselect(*bwmethod***)** specifies the bandwidth selection procedure to be used. Options are:
- **mserd** one common MSE-optimal bandwidth selector for the RD treatment effect estimator.
- **msetwo** two different MSE-optimal bandwidth selectors (below and above the cutoff) for the RD treatment effect estimator.
- **msesum** one common MSE-optimal bandwidth selector for the sum of regression estimates (as opposed to difference thereof).
- **msecomb1** for min(**mserd**,**msesum**).
- **msecomb2** for median(**msetwo**,**mserd**,**msesum**), for each side of the cutoff separately.
- **cerrd** one common CER-optimal bandwidth selector for the RD treatment effect estimator.
- **certwo** two different CER-optimal bandwidth selectors (below and above the cutoff) for the RD treatment effect estimator.
- **cersum** one common CER-optimal bandwidth selector for the sum of regression estimates (as opposed to difference thereof).
- **cercomb1** for min(**cerrd**,**cersum**).
- **cercomb2** for median(**certwo**,**cerrd**,**cersum**), for each side of the cutoff separately.

Note: MSE = Mean Square Error; CER = Coverage Error Rate.
Default is **bwselect(mserd)**.

bwjoint forces all bandwidths to be the same across groups. By default, separate bandwidths are computed for each group when **covs_hte()** is a factor or 0/1 variable; for continuous **covs_hte()** a single joint bandwidth is always used regardless of **bwjoint**.

Variance-Covariance Estimation

vce(*vcetype [clustervar]***)** specifies the variance-covariance matrix estimator. Without a cluster variable: **hc0**, **hc1**/**robust**, **hc2**, **hc3**. With a cluster variable: **vce(cr1** *clustvar***)** (default; standard cluster-robust sandwich, matching regress, vce(cluster ...)), **vce(cr2** *clustvar***)** (Bell-McCaffrey leverage-adjusted), or **vce(cr3** *clustvar***)** (block jackknife). Legacy: **vce(cluster** *clustvar***)** is an alias for **vce(cr1** *clustvar***)**; **vce(hc0/hc1** *clustvar***)** is remapped to **cr1**, **vce(hc2** *clustvar***)** to **cr2**, **vce(hc3** *clustvar***)** to **cr3** (with a warning).
Default is **vce(hc3)** when no cluster variable is supplied; **vce(cr1** *clustvar***)** when one is.

level(*#***)** specifies the confidence level (in percent) for confidence intervals. Default is **level(95)**.

Example:

Setup using Granzier, Pons, and Tricaud (2023) Data
. use rdhte_dataset.dta

RD-HTE Estimation by left/right groups
```
. rdhte y x, covs_hte(i.left)
```

RD-HTE Estimation by left/right groups with common bandwidth
```
. rdhte y x, covs_hte(i.left) bwjoint
```

RD-HTE Estimation by left/right groups and strong
```
. rdhte y x, covs_hte(i.left#i.strong)
```

RD-HTE Estimation using a continuous variable
```
. rdhte y x, covs_hte(c.mean_strength_nat1)
```

RD-HTE Estimation using a continuous variable with clustered standard errors
```
. rdhte y x, covs_hte(c.mean_strength_nat1) vce(cluster id_district)
```

Stored results

rdhte stores the following in **e()**:

Scalars

e(N)	sample size after listwise deletion on (y, x)
e(N_h_l)	effective sample size on the left of the cutoff (within bandwidth, summed across groups)
e(N_h_r)	effective sample size on the right of the cutoff (within bandwidth, summed across groups)
e(c)	cutoff value
e(p)	order of the polynomial used for the point estimator
e(q)	order of the polynomial used for bias correction
e(level)	confidence level

Macros

e(cmd)	**rdhte**
e(depvar)	name of dependent variable
e(runningvar)	name of running variable
e(clustvar)	name of cluster variable, when **vce(cluster ...)** or **vce(hc2 ...)** was specified
e(covs)	names of efficiency covariates from **covs_eff()**, when supplied
e(vce_select)	vcetype label (HC1/HC2/HC3/Cluster)
e(kernel)	kernel type used (Triangular / Epanechnikov / Uniform)
e(bwselect)	bandwidth selection procedure
e(precision)	storage precision used for generated working variables

Matrices

e(b)	bias-corrected estimates (1 x I row vector)
e(V)	variance-covariance matrix of **e(b)** (I x I)
e(h)	I x 2 matrix of left/right bandwidths, one row per group
e(tau_hat)	1 x I conventional p-order local-polynomial estimates
e(tau_bc)	1 x I bias-corrected local-polynomial estimates
e(tau_se)	I x 1 robust standard errors
e(tau_V)	I x I robust variance-covariance matrix
e(tau_t)	I x 1 robust z-statistics
e(tau_pv)	I x 1 robust p-values
e(tau_N)	I x 2 effective sample sizes per group, per side
e(tau_ci_lb)	I x 1 robust lower confidence-interval bounds
e(tau_ci_ub)	I x 1 robust upper confidence-interval bounds

References

Calonico, Cattaneo, Farrell, Palomba and Titiunik. 2025a. Treatment Effect Heterogeneity in Regression Discontinuity Designs. *Working Paper.*

Calonico, Cattaneo, Farrell, Palomba and Titiunik. 2025b. rdhte: Conditional Average Treatment Effects in RD Designs. *Working Paper.*

Granzier, Pons, and Tricaud. 2023. Coordination and Bandwagon Effects: How Past Rankings Shape the Behavior of Voters and Candidates. *American Economic Journal: Applied Economics*, 15(4): 177-217.

Cattaneo and Titiunik. 2022. Regression Discontinuity Designs. *Annual Review of Economics*, 14: 821-851.

Calonico, S., M. D. Cattaneo, and M. H. Farrell. 2020. Optimal Bandwidth Choice for Robust Bias Corrected Inference in Regression Discontinuity Designs. *Econometrics Journal*, 23(2): 192-210.

Calonico, Cattaneo, Farrell, and Titiunik. 2019. Regression Discontinuity Designs using Covariates. *Review of Economics and Statistics*, 101(3): 442-451.

Calonico, Cattaneo, and Titiunik. 2014. Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs. *Econometrica*, 82(6): 2295-2326.

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